



Diffuse glioma classification with deep learning and explainability: addressing challenges in histopathology image analysis

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Abstract

Brain tumors are among the most lethal diseases, causing significant neurological damage and high mortality. Diffuse gliomas, a highly invasive subtype, challenge treatment due to widespread brain infiltration and therapy resistance. This paper proposes a deep learning framework for detecting diffuse glioma and its subtypes from H&E-stained WSI patches, with both patch-level and patient-level detection, along with visualization to enhance explainability. Automation of detection enhances speed and consistency, reducing the workload of pathologists significantly. The proposed framework consists of three stages: a diffuse glioma patch classifier, a diffuse glioma subtype classifier, and a patient-level classifier. Additionally, four different explainable AI techniques including, Grad-CAM, Grad-CAM++, Score-CAM, and Smooth Grad-CAM++ were employed to highlight the diffuse glioma-affected areas. This study utilizes the DeepHisto dataset consisting of tiles from H&E-stained WSIs of 28 adult-type diffuse glioma cases from the NCP, Luxembourg National Health Laboratory. The data were preprocessed using a novel technique and trained with classification models including DenseNet121, MobileNetV2, an ensemble of DenseNet121 and MobileNetV2, and Vision Transformer. The ensemble model showed robust performance in diffuse glioma patch classification, achieving a specificity of 96.14% and an accuracy of 91.33%. The diffuse glioma subtype classifier recorded a specificity of 92.26%. For patient-level identification, the diffuse glioma classifier achieved an F1 score of 93.53%, while the diffuse glioma subtype classifier reached an accuracy of 95.23% and a specificity of 97.61%. The proposed deep learning framework for the automated detection of diffuse glioma and its subtypes demonstrated promising results. This framework has the potential to serve as a reliable diagnostic tool, enabling efficient, accurate, and cost-effective analysis of diffuse gliomas.

Keywords Adult-type diffuse glioma · Histopathology image analysis · H&E-stained WSIs · Deep learning-based classification · Explainable AI · CAM visualization · Brain tumor diagnosis

1 Introduction

The brain, as the central organ of the nervous system, processes sensory information and directs bodily actions through neurons. It coordinates all body activities by receiving, analyzing, and sending instructions to maintain function and control (Khan et al. 2022). Brain tumors are abnormal growths of cells within the brain or surrounding structures, disrupting normal neurological function (Sridhar et al. 2018). Brain tumors are broadly classified into primary brain tumors, originating within the brain, and secondary (metastatic) tumors, which spread from other parts of the body (Tandel et al. 2019). Brain tumors rank among the most fatal diseases globally, with a significant impact and high mortality rates, affecting people of all ages (Nazir et al. 2021; Sharif et al. 2021; Panda and Panda 2019; Kaur

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and Doegar 2019). In 2022, brain and central nervous system (CNS) cancers accounted for approximately 330,000 new cases and 251,000 deaths globally (Bray et al. 2024). A recent report, Cancer Statistics 2023, by the American Cancer Society, estimates that approximately 25,400 people in the U.S. will be diagnosed with a brain tumor, with an anticipated 18,000 related deaths in 2023 (Siegel et al. 2023). Primary brain tumors are categorized into two types: benign (non-cancerous) and malignant (cancerous), based on their growth patterns and potential to invade surrounding tissues. Benign tumors are characterized by slow growth, well-defined borders, and a lack of spread to other areas. In contrast, malignant tumors grow quickly, invade nearby tissues, and have the potential to metastasize to other parts of the body (Wang et al. 2019). Brain tumors are classified into four grades, reflecting their growth dynamics and malignant potential. Grades I and II are considered low-grade, characterized by slower growth and minimal aggressiveness. In contrast, grades III and IV are high-grade, marked by rapid proliferation and a higher propensity for metastasis (Johnson et al. 2017; Amin et al. 2022; Alwazzan et al. 2023). The evolution of brain tumors generally manifests within these established grades, influenced by complex genetic and environmental interactions, which are the focus of extensive ongoing investigations. Among primary brain tumors, gliomas are the most common malignant type, arising from glial cells that support neuronal function. Diffuse gliomas exhibit significant heterogeneity at molecular, histological, and imaging levels, posing challenges in diagnosis and treatment (Aum et al. 2014; Lemée et al. 2015; Rathore et al. 2020). The 2021 World Health Organization (WHO) classification emphasizes molecular features for diffuse glioma categorization, refining diagnostic precision. Adult-type diffuse gliomas are classified as astrocytoma (Astro), IDH-mutant and lacking 1p/19q co-deletion; oligodendroglioma (Oligo), IDH-mutant and 1p/19q-codeleted, which typically has a favorable prognosis; and glioblastoma (Glio), IDH-wildtype, the most aggressive subtype with frequent genetic alterations (Louis et al. 2021). Glioblastoma limits overall survival to around 15 months, even with standard treatments that include maximal surgical resection, radiotherapy, and chemotherapy with temozolomide (Marenco-Hillebrand et al. 2020). Its aggressive nature and poor prognosis result in less than 5% of patients surviving beyond five years (Lapointe et al. 2018). The causes of most brain tumors remain unclear; however, a comprehensive medical history can aid in identifying potential risks or contributing factors for their development (Sharif et al. 2020). High-risk groups include older adults, males, and individuals with a family history of brain tumors (Ostrom et al. 2015). Without prompt intervention, a brain tumor may disrupt normal brain functions and lead to serious health risks. Current clinical

techniques for identifying brain tumors include computed tomography (CT) and magnetic resonance imaging (MRI) scans, followed by biopsy and histopathologic and molecular diagnostics. MRI, particularly with functional and diffusion tensor imaging, is widely used for detailed structural and functional assessments, while CT is preferred for rapid imaging in emergencies (Jin et al. 2020). Biopsy, often guided by stereotactic methods, is the definitive procedure for obtaining tumor samples to determine type and grade (Madlener and Gojo 2020). Histopathology image analysis, performed on these biopsy samples, remains the gold standard for accurate cancer identification and diagnosis (Veta et al. 2014; Xu et al. 2017). Molecular diagnostics and immunohistochemistry (IHC) further enhance tumor characterization. Molecular techniques identify key markers like IDH mutations and 1p/19q codeletion, while IHC detects protein expression such as GFAP and Ki-67, aiding in precise diagnosis and treatment planning (Takano et al. 2016; Nishikawa et al. 2022). However, these techniques can be time-consuming and heavily depend on the expertise of clinicians and pathologists, which may lead to delays in diagnosis. Early and accurate diagnosis of brain tumors is crucial for guiding timely interventions, such as surgery or targeted therapies, which directly improve treatment outcomes, survival rates, and patient quality of life (Sobhaninia et al. 2020; Majib et al. 2021). Artificial Intelligence (AI) and Machine Learning (ML) have become key tools in improving brain tumor diagnosis. Recent research has increasingly focused on utilizing these technologies to improve the speed and accuracy of brain tumor detection. Recent advancements have focused on MRI for brain tumor diagnosis, while the role of histopathology image analysis has been less explored. Despite this, histopathology remains the gold standard for tumor and cancer diagnosis (Xu et al. 2017), emphasizing the importance of integrating pathology data with AI and ML to enhance diagnostic accuracy.

Elazab et al. (Elazab et al. 2024) proposed a hybrid YOLOv5–ResNet50 model for detecting and grading gliomas on histopathology WSIs, achieving 97.2% accuracy and a Dice coefficient of 97% on the TCGA dataset. While their approach effectively distinguishes between low-grade and high-grade gliomas, it does not specifically address the classification of diffuse glioma subtypes as defined by the 2021 WHO classification. Jin et al. (Jin et al. 2021) proposed a deep learning model using a DenseNet architecture with a squeeze-and-excitation block (SD-Net_WCE) for classifying glioma subtypes, trained on over 79,990 images from 267 patients. Despite the large dataset, the model achieved a patch-level accuracy of 86.5% and patient-level accuracy of 87.5%, indicating that while the approach is innovative, there is potential for improvement in achieving higher diagnostic accuracy. Kurc et al. (Kurc et al. 2020) conducted

a study analyzing glioma subtypes using datasets from the 2018 CPM challenge, which included 32 training cases and 20 test cases sourced from TCGA1 and TCIA2. Their methodology combined CNN-based radiology for segmenting tumors from imaging data with pathology analysis, which processed whole-slide images into smaller patches for detailed examination. This dual approach sought to integrate radiographic and histologic data to improve diagnostic accuracy. However, the study primarily focused on patch-level accuracy, reporting an accuracy of 75%, 80%, and 90% for different methods. The research was constrained by its emphasis on image patches rather than comprehensive patient-level classification, limiting the clinical relevance of the findings for individual patient diagnosis. Wang et al. (Wang et al. 2023) analyzed glioma subtypes using three datasets comprising a total of 2624 patients, categorized into six glioma classes. Their approach combined ResNet-50-based feature extraction with K-means clustering for patch-level classification, followed by majority voting for patient-level classification. Binary classification achieved an accuracy of 94.5% for astrocytoma and 86.5% for oligodendroglioma, while multiclass classification yielded an accuracy of 95% for glioblastoma multiforme (GBM), but lower performance for Oligodendroglioma, grade III (OIII) (57%). Despite promising results, variability in subclass performance, such as OIII (57%), highlights limitations in the model's ability to generalize across glioma subtypes. Hewitt et al. (Hewitt et al. 2023) classified brain tumor subtypes using a dataset of 2845 patients from three cohorts (1882, 864, 99), categorized into Astro, Oligo, and Glio classes. Their approach utilized an attention-based multiple-instance learning (attMIL) model, extracting features from patches of WSIs for training and prediction. The study reported an accuracy of 81% for Oligo in direct prediction, which improved to 83% with stacked sequential prediction. However, the model assumes that all patches share the WSI's class label, overlooking potential variability between patches. Additionally, the absence of visualization techniques limits both the interpretability of the model and its applicability in clinical settings. Building on prior research employing methods such as DenseNet architectures, CNN-based radiology-pathology integrations, ResNet-50 feature extraction, and attention-based MIL approaches, we sought to address key gaps in diffuse glioma classification, including limited visualization techniques and variability between patch-level and patient-level analyses. While previous studies often relied on single CNN models, our approach highlights the potential of ensemble CNNs to enhance performance by combining complementary features and reducing model bias.

In our research, we utilized a publicly available histopathology dataset of diffuse glioma to develop and evaluate

a robust classification framework. To enhance readability and avoid repetitive use of the phrase “diffuse glioma,” we have occasionally used the term “glioma” where the context implies the diffuse subtype. Histopathology images often exhibit staining variations, and in some cases, it can be challenging for the model to extract meaningful features from raw images (Wu et al. 2022). To address this, we implemented a novel image normalization technique to standardize image appearance and improve feature extraction. We performed binary classification to distinguish between diffuse glioma and normal brain tissue, followed by diffuse glioma subtype classification into astrocytoma, glioblastoma, and oligodendroglioma. Our approach leveraged an ensemble of two CNN-based deep learning models, enhancing accuracy and robustness. Importantly, we addressed both patch-based and patient-based classification for a comprehensive analysis. Additionally, we employed various Class Activation Mapping (CAM) techniques to visualize regions contributing to model predictions, improving interpretability (Huff et al. 2021) and clinical relevance. To highlight the key contributions of our study are:

- Introduced a novel image normalization technique to address staining variations in histopathology images, improving feature extraction.
- Performed binary classification (diffuse glioma vs. normal brain tissue) and diffuse glioma subtype classification (astrocytoma, glioblastoma, and oligodendroglioma) using both patch-based and patient-based approaches for comprehensive diagnostic analysis.
- Leveraged an ensemble of two CNN-based deep learning models, enhancing classification accuracy and robustness.
- Enhanced model interpretability and clinical relevance through the use of Explainable AI (XAI) techniques, such as Class Activation Mapping (CAM) visualizations.

2 Materials and methods

2.1 Dataset description

In our study, we utilized the publicly available DeepHisto dataset (Mittelbronn et al. 2023), which comprises tiles extracted from hematoxylin and eosin (H&E)-stained Whole Slide Images (WSIs) of 28 adult-type diffuse glioma cases collected at the National Center of Pathology (NCP), Luxembourg National Health Laboratory (LNS) between 2017 and 2021. WSIs were scanned using an IntelliSite Ultra-Fast digital slide scanner at a resolution of 0.25 $\mu\text{m}/\text{pixel}$. The dataset contains five classes: astrocytoma, glioblastoma, oligodendroglioma, normal brain (white and gray matter),

Fig. 1 Representative Sample Images from the Experimental Dataset

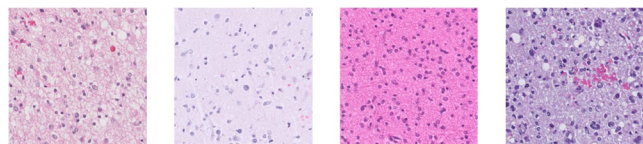
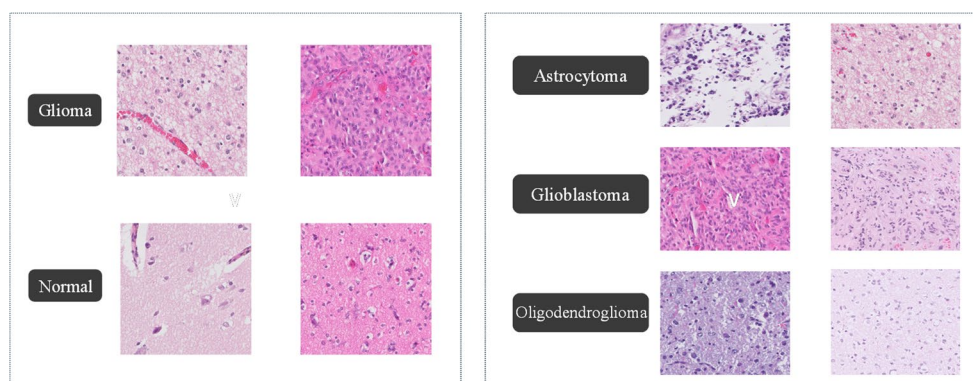


Fig. 2 Staining Variability in the Astrocytoma Class

and necrosis. Glioma subtypes are classified as IDH-mutant 1p/19q-codeleted oligodendroglioma, IDH-mutant astrocytoma, and IDH-wildtype glioblastoma, per the 5th edition of the WHO classification of central nervous system tumors (Louis et al. 2021). Regions of interest within the WSIs were annotated by a board-certified pathologist and divided into 512×512 tiles, each labeled with a specific class. The dataset is divided into two subsets: a training set containing 38,544 patches from 25 patients and a test set comprising 4,174 patches from 3 patients. For this study, the focus was exclusively on the training set, excluding the necrosis class, resulting in a total of 38,155 patches used for experiments. Among the 25 patients, 21 had glioma (8 astrocytoma, 7 glioblastoma, 5 oligodendroglioma) and only 4 were normal brain samples. Although glioma cases yield about 8,000 patches in total, the normal class contributes roughly 29,000 patches, over 25,000 of which come from a single patient. Figure 1 presents representative images from our experimental dataset.

2.2 Data preprocessing

In this study, our goal is to classify diffuse glioma and its subtypes from H&E-stained WSI patches. In histopathological image analysis, the heterogeneity in staining and characteristics observed in WSIs poses significant challenges, often complicating the process of pattern recognition (Raza et al. 2024; Runz et al. 2021). To address this, various image normalization techniques have been developed; however, none proved effective on our dataset. Our dataset exhibits substantial staining variability, and an example is shown in Fig. 2 for the astrocytoma class. In this study, we implemented a distinctive image normalization approach

to address staining variations present in the dataset. The approach involved a linear combination of two established normalization techniques: the Reinhard and Macenko. The method generated new images by taking a weighted sum of the images normalized by each technique, effectively combining the characteristics of both methods. The linearly blended images were split subject-wise into training, testing, and validation sets based on five-fold cross-validation, ensuring an 80–20 split between training and testing. Additionally, 10% of the training data was reserved for validation purposes. The split ensured that data from any single patient was included in only one set to prevent data leakage (Chowdhury et al. 2023). As the number of images varied across classes, we employed image augmentation techniques to balance the dataset (Chowdhury et al. 2020). Reinhard- and Macenko-normalized images (Macenko et al. 2009; Reinhard et al. 2001) were used as augmented data to achieve this balance (Shen et al. 2022). Figure 3 demonstrates examples of images processed with different normalization techniques. This technique has proven to outperform models trained on raw data, those using only single normalization techniques, or those relying on general augmentations such as rotation and flipping.

2.3 Proposed method

In this study, we propose an automated framework for the classification and identification of diffuse gliomas in H&E-stained Whole Slide Images (WSIs). The framework consists of three main stages: a diffuse glioma patch classifier, a diffuse glioma subtype classifier, and a patient-level classifier. Additionally, XAI techniques are used to generate heatmaps for localizing tumor regions. To address staining variations in the dataset, a preprocessing technique is applied, combining Reinhard and Macenko normalization methods through a weighted linear blending approach.

In the first stage, patches extracted from WSIs are classified as either glioma or normal tissue. In the second stage, glioma patches identified in the previous step are further categorized into three subtypes: astrocytoma, glioblastoma,

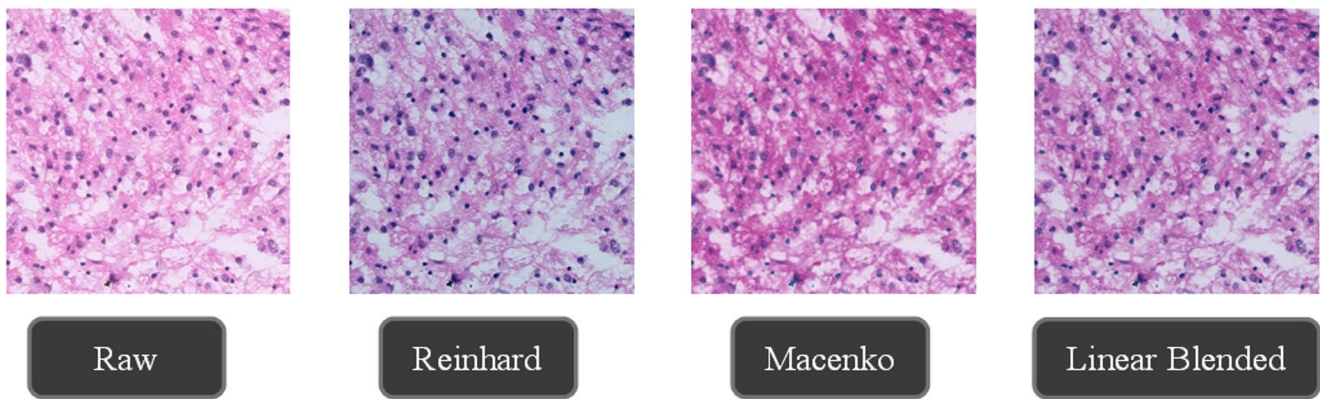


Fig. 3 Visualization of Different Normalization Techniques

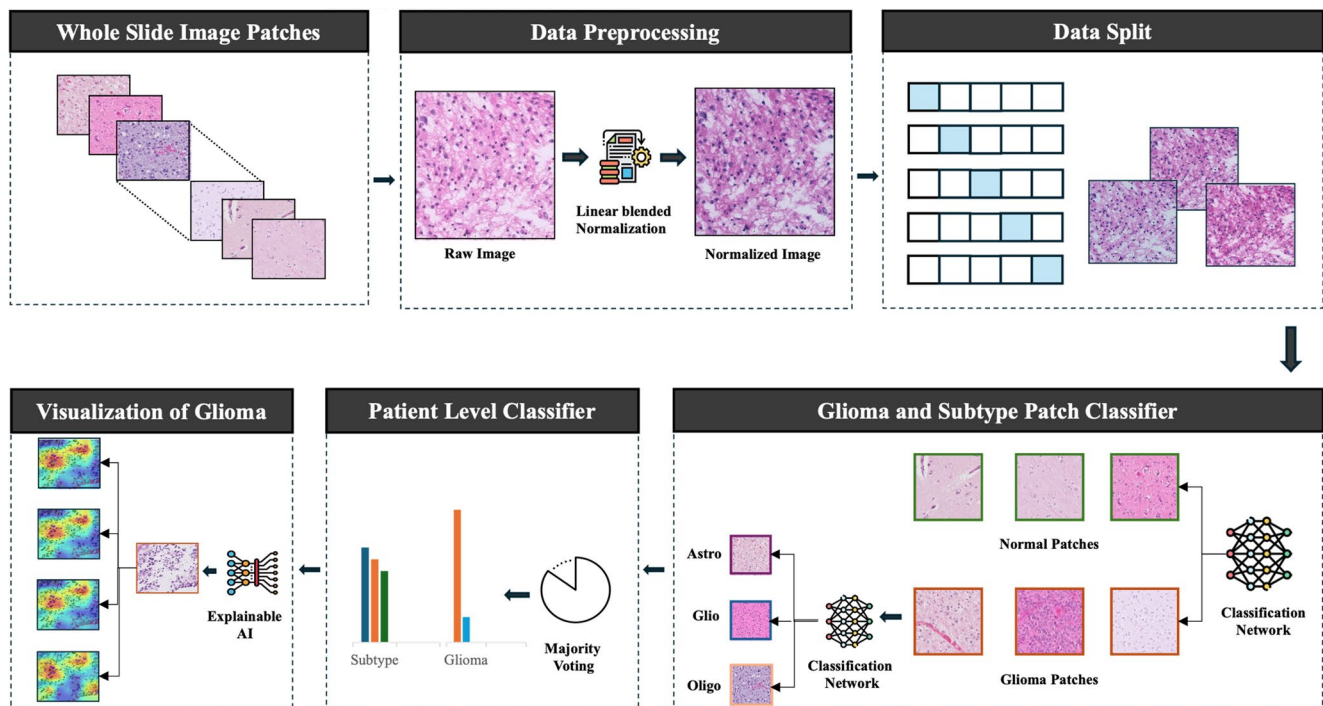


Fig. 4 Proposed Automated Framework for Diffuse Glioma Classification and Localization

and oligodendroglioma. A CNN ensemble model is developed for classification tasks. For the patient-level classifier, majority voting is used to determine both the presence of glioma and its subtype for individual patients. To improve interpretability, visualization techniques such as Grad-CAM, Grad-CAM++, SmoothGrad-CAM++, and Score-CAM are used to generate heatmaps, highlighting the regions identified by the model. Figure 4 illustrates the workflow of the proposed framework.

2.3.1 Diffuse glioma patch classifier

In the initial phase of our study, we developed an automated system to differentiate between normal and glioma patches

in WSIs, a distinction crucial for accurate diagnosis and effective treatment planning. This system employs deep learning architectures, including CNN and Vision Transformer (ViT), as the foundational framework for classification models.

CNNs are extensively used in image processing for their ability to capture spatial hierarchies, particularly beneficial in medical imaging to identify nuanced patterns crucial for accurate diagnosis (Krizhevsky et al. 2017). By applying multiple filter layers, CNNs extract hierarchical features at different abstraction levels. Ensembling different CNN architectures can enhance model performance by combining their outputs or features, providing a comprehensive data representation. This approach is advantageous in medical

image classification, improving prediction accuracy by leveraging detailed feature analysis (Nguyen et al. 2021; Nanni et al. 2021; Kumar et al. 2016; Pranto and Jahan, 2024). While CNNs have long been a staple in image analysis tasks due to their effectiveness in handling spatial hierarchies, newer architectures like ViT (Dosovitskiy 2020) and Data-Efficient Image Transformers (DeiT) (Touvron et al. 2021) offer alternative approaches. ViTs utilize self-attention mechanisms, treating images as sequences of patches, which helps capture long-range dependencies within the image data. DeiT, a variation of ViT, improves training efficiency by employing a distillation token, enabling it to learn effectively from smaller datasets without external pretraining. The performance of models such as CNNs, ViTs, and DeITs depends on the characteristics and complexity of the dataset, as different models are optimized to handle specific types of data structures and patterns (Granizo et al. 2024). In this research, we systematically evaluated multiple deep learning models for classification tasks. The classification performance of the four best-performing models—DenseNet121 (Huang et al. 2017), MobileNetV2 (Howard et al. 2018), an ensemble of DenseNet121 and MobileNetV2, and a Vision Transformer-based approach—was comprehensively presented. DenseNet121 (Huang et al. 2017) is a convolutional neural network that utilizes dense connections between layers, ensuring maximum feature reuse and efficient gradient flow. Its compact architecture reduces the number of parameters while maintaining high performance, making it effective for complex classification tasks. MobileNetV2 (Howard et al. 2018) is a lightweight CNN designed for mobile and embedded applications, leveraging inverted residuals and linear bottlenecks. It achieves remarkable

efficiency with minimal computational overhead, making it ideal for scenarios with limited resources.

Our proposed ensemble model integrates DenseNet121 and MobileNetV2 architectures for feature extraction. DenseNet121 features are extracted up to the third Dense Block, while MobileNetV2 features are extracted up to the last ConvBNReLU layer. To ensure spatial alignment, an upsampling layer is applied to the MobileNetV2 outputs before concatenation with DenseNet121 features. The concatenated features are processed through fully connected layers for final classification, with auxiliary classifiers providing additional outputs from each backbone. While ViT architectures are effective in capturing long-range dependencies and global contextual information, their performance is not uniformly optimal across all datasets (Zhu et al. 2023; Lu et al. 2022). In this study, the ensemble and standalone CNN models exhibited better performance than the ViT architecture on the given dataset. This observation underscores the variability in model performance depending on the dataset characteristics. Figure 5 depicts the architecture of the proposed ensemble model.

2.3.2 Diffuse glioma subtype classifier

Accurate identification of diffuse glioma subtypes is crucial for determining appropriate treatment strategies and improving patient outcomes (Kan et al. 2020). At this stage, we focused on identifying the subtype of diffuse glioma. The methodology employed here aligns with the approach detailed in Sect. 2.3.1, *Diffuse Glioma Patch Classifier*. To assess the impact of data preprocessing on detection performance, we conducted analyses across four distinct scenarios: Raw (without normalization), Reinhard normalization,

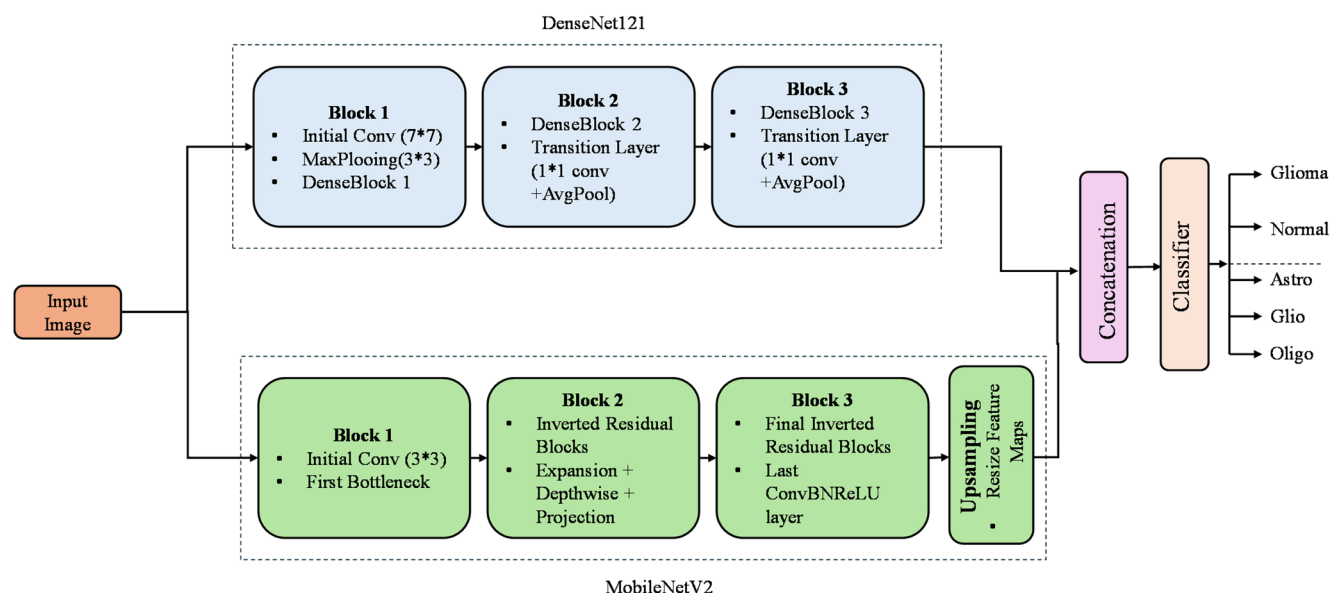


Fig. 5 Architecture of the proposed Ensemble Model

Macenko normalization, and our novel linear blended normalization approach. For single normalization techniques, each dataset was uniformly processed using the specified method before splitting. For the first three scenarios, we augmented the training set by employing traditional techniques such as random rotations, horizontal and vertical flips, and zooming. The results revealed that linear blended normalization significantly improved detection accuracy compared to the other approaches, demonstrating its effectiveness in enhancing subtype classification.

2.3.3 Patient-Level classifier

In this stage, we performed patient-wise classification to identify diffuse glioma and its subtypes for each patient, a critical step in aligning diagnostic outcomes with personalized treatment plans (Wang et al. 2024). We employed a majority voting approach on the patch-based results, where the majority prediction from patches of a given patient (i.e., predictions exceeding a 50% threshold) determined the patient's classification (Faisal et al. 2023; Jahan et al. 2025). This method addressed both glioma detection and subtype identification, providing a clinically relevant framework.

2.3.4 Visualization of diffuse glioma patches

Visualization plays a vital role in understanding and interpreting the behavior of machine learning models, particularly in domains like histopathology (Sureka et al. 2020; Vellido 2020). Diffuse gliomas present unique challenges that require interpretable methods to ensure model predictions align with clinically relevant features and support informed decision-making (Fang et al. 2023). This section outlines the visualization of diffuse glioma patches using generalized heatmaps and four XAI techniques: Grad-CAM (Selvaraju et al. 2020), Grad-CAM++ (Chattopadhyay et al. 2018), Smooth Grad-CAM++ (Omeiza et al. 2019), and Score-CAM (Wang et al. 2020). These methods provide insights into the areas of the images that are most significant to the model's detection process, enabling a detailed examination of its interpretative focus. Grad-CAM (Selvaraju et al. 2020) serves as a baseline for identifying regions of interest, Grad-CAM++ (Chattopadhyay et al. 2018) enhances localization accuracy, Smooth Grad-CAM++ (Omeiza et al. 2019) reduces noise for clearer and more focused visualizations, and Score-CAM (Wang et al. 2020) provides gradient-independent sharper results. Presenting all four techniques allows for a comprehensive comparison, offering clarity on the model's decision-making framework and ensuring robust interpretability in the context of glioma identification.

2.3.5 Training and testing methodology

The dataset, comprising data from 25 patients, was divided into five folds using a patient-wise cross-validation approach. For each fold, one subset was designated as the test set, while the remaining subsets were utilized for training, with a portion of the training dataset set aside for validation. This approach provided a structured framework for evaluating both classification tasks effectively. In the diffuse glioma patch classifier, the dataset was imbalanced, with 21 patients diagnosed with diffuse glioma and only 4 classified as normal. A single normal patient contributed over 25,000 patches, reflecting the skewed distribution. In comparison, the diffuse glioma subtype classification task was more balanced, providing a more even distribution of data across the classes. Patches measuring 512×512 pixels at 20x magnification were resized to 224×224 pixels to match the input dimensions required for ImageNet-pretrained weights. Classification networks were trained with a batch size of 16 over a maximum of 100 epochs, with an early stopping criterion applied after 10 epochs if no improvement was observed. Early stopping was consistently applied across all models during training to prevent overfitting. The best-performing model from each training cycle was saved based on validation set performance. Dropout layers in the final classifier and weight decay regularization (L2 penalty with weight decay=0.0001) via the optimizer were also utilized to further mitigate overfitting. Preliminary experiments with various learning rates led to the selection of a learning rate of 0.0001 with the Adam optimizer. For classification tasks, the Negative Log Likelihood Loss (NLLLoss) function was used to evaluate the model's performance and guide optimization.

All experiments were conducted on a hardware setup consisting of an NVIDIA GeForce RTX 4090 GPU and an AMD Radeon™ graphics card. The system featured 128 GB of RAM and was driven by an AMD Ryzen 9 7950×3D 16-core processor. All training and model evaluations were carried out using Visual Studio Code.

2.4 Evaluation matrices

To evaluate the performance of the classifiers, several metrics were employed, all derived from the confusion matrix. The confusion matrix provides four critical values: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). These values form the basis for computing evaluation metrics such as Accuracy (A), Recall or Sensitivity (R), Precision (P), Specificity (S), and F1-Score (F1) (Sokolova and Lapalme 2009). The mathematical expressions for these five metrics are presented in Eqs. (1)–(5). Metrics were computed separately for

each class to analyze the classifier's ability to distinguish between categories. As the number of instances varied across classes, class-wise results alone could be misleading without accounting for this imbalance. A weighted average of each metric, reflecting class distribution, was calculated to ensure balanced evaluation and address class imbalances, as detailed in Eq. (6).

$$A = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$P = \frac{TP}{TP + FP} \quad (2)$$

$$R = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = 2 \frac{Precision \times Sensitivity}{Precision + Sensitivity} \quad (4)$$

$$S = \frac{TN}{TN + FP} \quad (5)$$

$$Wighted\ Average\ Metric = \sum_{i=1}^n \frac{N_i}{N} \times Metric_i \quad (6)$$

The equation uses the following parameters: N_i the number of instances in class i ; N , the total number of instances; n , the number of classes; $Metric_i$ (e.g., precision or recall for class i), scaled by class proportions to compute weighted averages like precision. Additionally, Receiver Operating Characteristic (ROC) curves were plotted, and the Area Under the Curve (AUC) was calculated to assess the classifiers' ability to distinguish between positive and negative instances. Metrics were computed using five-fold cross-validation, with predictions concatenated for final analysis to ensure a comprehensive evaluation across all classes and folds. These metrics were computed separately at both the

patch level and patient level to provide a thorough assessment of the model's performance.

3 Results

3.1 Diffuse glioma patch classifier's performance

This section presents a performance comparison of the Diffuse Glioma Patch Classifier using four deep learning models: DenseNet121, MobileNetV2, their ensemble, and Vision Transformer. The results are presented in Table 1, and the confusion matrices are shown in Fig. 6. DenseNet121 exhibited consistent performance across both glioma and normal classes, achieving an accuracy of 90.3%, a precision of 92.43%, an F1-score of 90.77%, and a specificity of 94.04%. In comparison, MobileNetV2 achieved slightly higher results, with an accuracy of 90.82%, an F1-score of 91.22%, and a specificity of 93.38%. MobileNetV2 demonstrated better performance in identifying normal patches, whereas DenseNet121 showed superior classification in the glioma class. The ViT model showed improved performance in the glioma class, with only 285 misclassified patches and surpassing the performance of DenseNet121 and MobileNetV2 in this class but underperforming in the normal class compared to the other models. ViT achieved an accuracy of 88.84%, a precision of 91.79%, and a specificity of 94.33%, demonstrating its effectiveness in correctly identifying diffuse glioma patches. The ensemble model, combining DenseNet121 and MobileNetV2, outperformed all other models across all evaluation metrics. It achieved an accuracy and a recall of 91.33%, a precision of 93.43%, an F1-score of 91.75%, and a specificity of 96.14%, highlighting its reliability in identifying glioma patches. To assess the significance of performance differences, paired t -tests were conducted on the final performance results. The ensemble model showed statistically significant improvements over

Table 1 Performance comparison of different models and classifiers

Classifier	Model/Approach	A (%)	P (%)	R (%)	F1 (%)	S (%)	p -value vs. Ensemble	Significance
Diffuse Glioma Patch Classifier	ViT	88.64	91.79	88.64	89.29	94.33	<0.0001	✓
	DenseNet121	90.30	92.43	90.31	90.77	94.04	<0.0001	✓
	MobileNetV2	90.82	92.54	90.82	91.22	93.38	0.0002	✓
	Ensemble	91.33	93.43	91.33	91.75	96.14	--	--
Diffuse Glioma Subtype Classifier	Raw	58.99	48.65	46.65	47.34	64.81	<0.0001	✓
	Macenko	66.65	55.66	54.53	52.04	75.90	<0.0001	✓
	Reinhard	71.47	61.09	57.36	56.64	81.09	<0.0001	✓
	ViT	82.77	78.41	75.01	75.48	87.98	<0.0001	✓
	MobileNetV2	88.02	83.09	82.47	82.63	91.15	0.0003	✓
	DenseNet121	90.11	86.46	85.59	85.57	91.31	0.0006	✓
	Ensemble	89.75	85.66	85.08	85.18	92.26	--	--
	Ensemble	89.75	85.66	85.08	85.18	92.26	--	--
Patient-Level Classifier	Diffuse Glioma Classifier	88.00	100	88.00	93.53	100	--	--
	Subtype Classifier	95.23	95.83	95.23	95.18	97.62	--	--

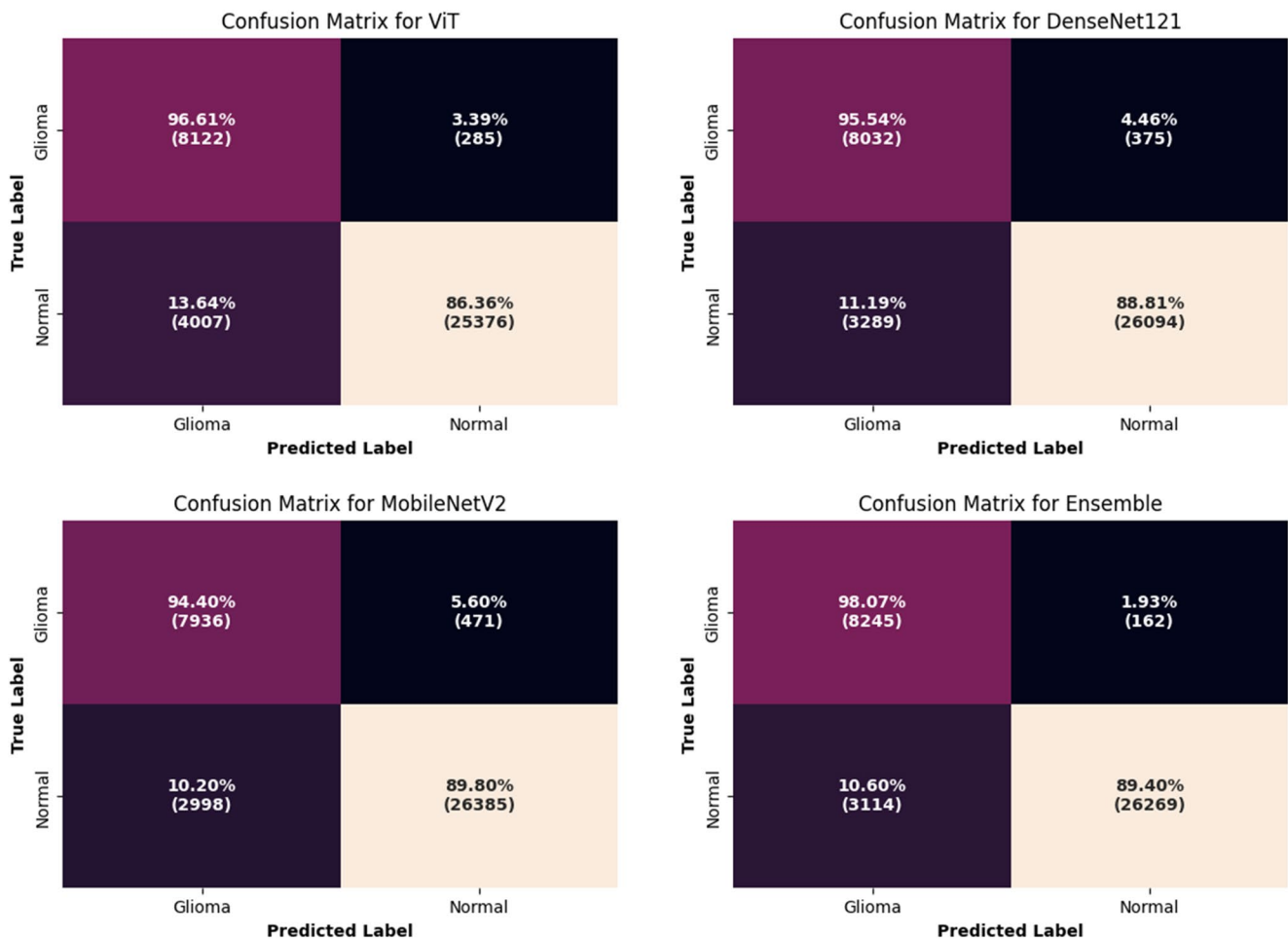


Fig. 6 Confusion Matrix of the Diffuse Glioma Patch Classifier

ViT ($p < 0.0001$), DenseNet121 ($p < 0.0001$), and MobileNetV2 ($p = 0.0002$), confirming its superior performance.

The confusion matrix indicated that the ensemble model had the lowest false-negative rate among all four models, reflecting the lowest likelihood of missing glioma cases. The slightly higher misclassification rate in the normal class may stem from the dataset's imbalance, as the greater proportion of normal patches compared to glioma patches biases the model's performance and influences the evaluation metrics. Furthermore, over 25,000 patches originated from a single normal patient, resulting in a skewed distribution and reduced variability. Figure 7 (a) illustrates the ROC plot along with the respective AUC values for the glioma patch classifier. The macro-average ROC curve achieves an AUC of 0.96, while the class-specific ROC curves for glioma and normal classifications also yield an AUC of 0.96.

3.2 Diffuse glioma subtype classifier's performance

This section begins by presenting the results of the Diffuse Glioma Subtype Classifier using our proposed ensemble

model under different preprocessing conditions: Raw (without normalization), Reinhard image normalization, and Macenko image normalization. These baseline results serve as a reference for evaluating the impact of preprocessing techniques. Following this, the performance of the classifier is presented under the proposed novel data preprocessing technique, highlighting its significance. The results of the four best-performing deep learning models including, DenseNet121, MobileNetV2, their ensemble, and ViT are discussed to demonstrate the effectiveness of our proposed novel preprocessing approach. The results are detailed in Table 1, while the confusion matrices are illustrated in Figs. 8 and 9.

In the initial scenario without normalization, detection performance yielded an accuracy of 58.99%, a precision of 48.65%, and a specificity of 64.81%. A significant portion of astrocytoma cases were incorrectly classified as oligodendroglioma, and similarly, many oligodendroglioma cases were misidentified as astrocytoma. Applying Macenko normalization resulted in an accuracy of 66% and a specificity of 75.90%, marking an improvement of more than 10%

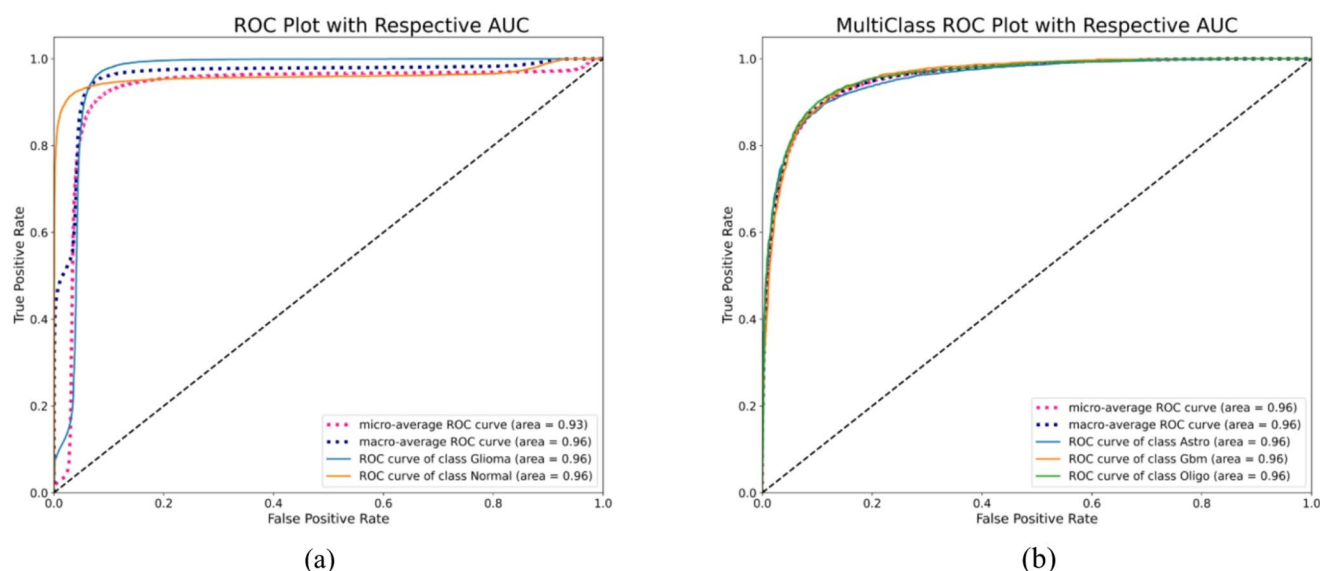


Fig. 7 (a) ROC Curve for the Diffuse Glioma Patch Classifier (b) ROC Curve for the Diffuse Glioma Subtype Classifier

compared to the unnormalized data. Although this method reduced the misclassification of oligodendroglioma cases, it led to a significant increase in mispredictions of astrocytoma cases. Using Reinhard normalization, we achieved an accuracy of 71.47%, a recall of 57.36%, and a specificity of 81.09%. However, this method resulted in a considerable increase in the misclassification of astrocytoma cases as glioblastoma, introducing a new diagnostic challenge. Thus, none of these techniques have provided a generalized model with sufficiently reliable performance for medical applications. For diffuse glioma subtype classification on our proposed preprocessed dataset, DenseNet121 achieved an accuracy of 90.11%, a precision of 86.46%, and a specificity of 91.31%. The model performed particularly well in identifying Astrocytoma cases compared to other subtypes. In contrast, MobileNetV2 showed slightly lower performance, with an accuracy of 88.02%, an F1-score of 82.63%, and a recall of 82.47%, excelling in the classification of Oligodendroglioma cases. ViT demonstrated superior performance in identifying glioblastoma cases; however, it misclassified the other two subtypes significantly, leading to reduced overall performance. ViT achieved an accuracy of 82.77%, a precision of 78.41%, an F1-score of 75.48%, and a specificity of 87.98%. The proposed ensemble model, however, delivered balanced performance across all subtypes. It achieved the highest specificity among all models at 92.26%, alongside an accuracy of 89.75% and a precision of 85.66%. Although the accuracy of DenseNet121 was slightly higher, this can be attributed to the relatively higher number of astrocytoma cases in the dataset, which had a greater influence on the performance metrics. These results demonstrate that our proposed novel data preprocessing technique significantly enhances the generalizability of the model's performance.

Analysis of the final results revealed statistically significant improvements by the ensemble model over individual architectures and preprocessing techniques ($p < 0.05$), reinforcing its effectiveness in subtype classification. Figure 7 (b) presents the ROC plot with respective AUC values for the glioma subtype classifier. Both the micro-average and macro-average ROC curves achieve an AUC of 0.96. Furthermore, all three classes—astrocytoma, oligodendroglioma, and glioblastoma—achieve an AUROC of 0.96, demonstrating the robustness and effectiveness of the proposed model.

3.3 Patient-Level classifier's performance

In this section, we present the results of the patient-level classification for both the diffuse glioma classifier and its subtype classifier, focusing on the performance of our proposed ensemble model. A patient is classified into a specific type if the majority of their patches are assigned to that type. The results are summarized in Table 1, with the corresponding confusion matrices displayed in Fig. 10 (a) Diffuse Glioma Classifier and (b) Diffuse Glioma Subtype Classifier.

For the patient-level diffuse glioma classifier, the model achieved an F1-score of 93.53%. Given the significant imbalance in the dataset, with a 21:4 ratio of glioma to normal patients, the F1-score serves as a reliable metric to assess performance under such conditions. The model showed promising results with only three misclassifications in the glioma class and none in the normal class, indicating its effectiveness in addressing the dataset imbalance. For diffuse glioma subtype classification, the model achieved a specificity of 97.61%, an accuracy of 95.23%, a precision of 95.83%, and an F1-score of 95.18%, with only one

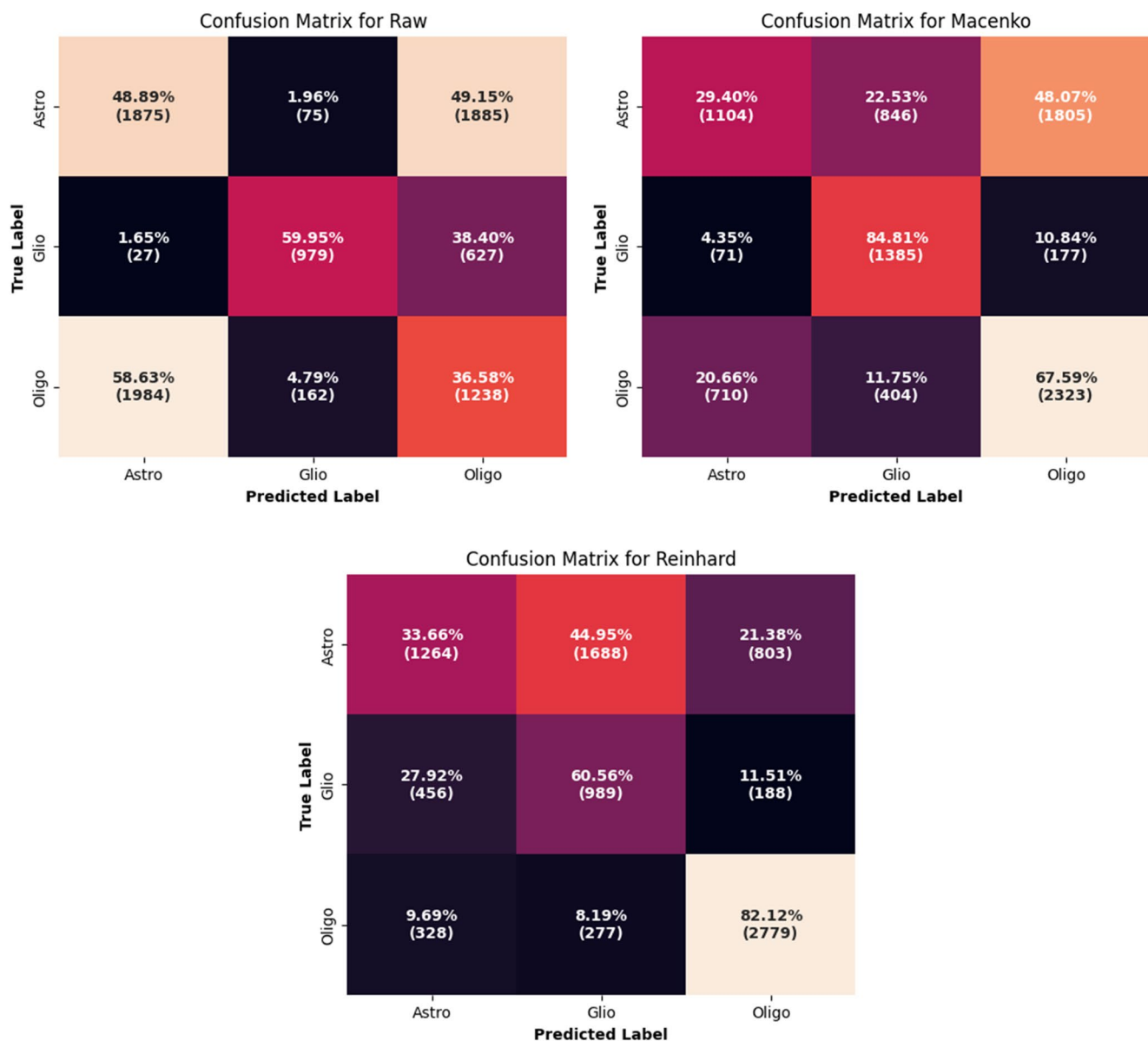


Fig. 8 Confusion Matrix of the Diffuse Glioma Subtype classifier (baseline)

misclassification in an oligodendroglioma case and none in astrocytoma or glioblastoma cases. These results reflect the model's ability to accurately distinguish between subtypes with minimal errors.

3.4 Visualization of diffuse glioma patches

In this stage, we utilized Grad-CAM, Grad-CAM++, SmoothGrad-CAM, and Score-CAM to generate heatmaps, visualizing the model's focus areas and the regions responsible for diffuse glioma identification. Diffuse glioma patches from our dataset were analyzed, and the resulting heatmaps consistently highlighted regions corresponding to glioma areas. These visualizations were validated by

pathologists, confirming a strong correspondence between the highlighted regions and the underlying histopathological features. The convergence of these techniques in marking similar regions underscores the model's explainability and demonstrates its reliability in identifying clinically relevant features. Figure 11 presents the visualizations of glioma patches generated using the XAI techniques.

4 Discussion

This study introduces an automated framework for the detection of diffuse glioma and its subtypes from H&E-stained whole slide images (WSIs). However, this comes

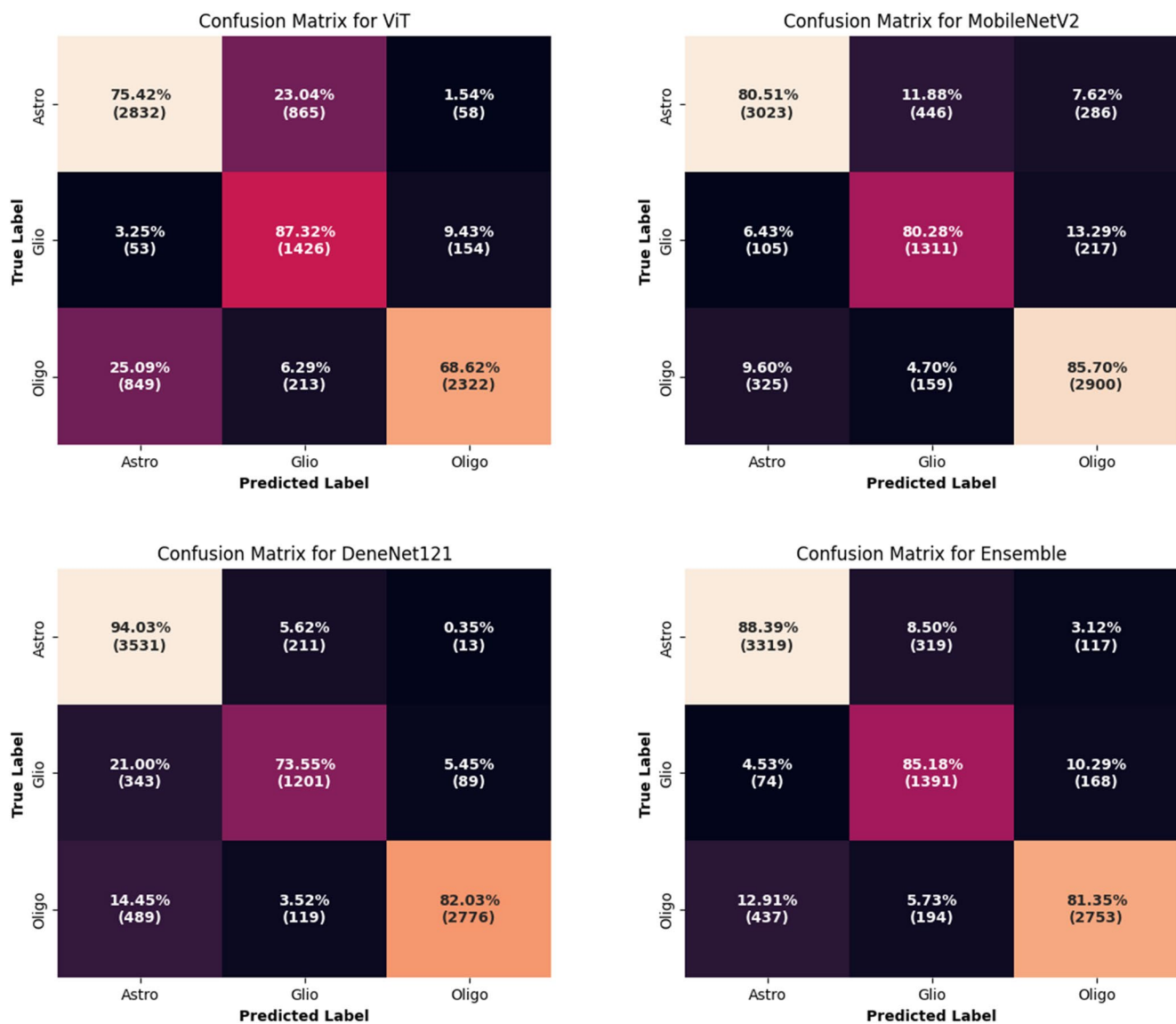


Fig. 9 Confusion matrix of the Diffuse Glioma Subtype Classifier with the Proposed Preprocessing technique

with challenges, as data variability and staining quality play a significant role in the model's performance. We proposed a novel data preprocessing technique to address such adversities, demonstrating proven improvements. Our study is structured into three key stages: diffuse glioma patch classifier, diffuse glioma subtype classifier, and patient-level classification. We developed a fully automated system designed to assist in diagnosing diffuse glioma in medical patients.

The framework begins by identifying patches with diffuse glioma through a classification stage, which showed promising performance across four deep learning models. MobileNetV2 achieved an accuracy of 90.82%, surpassing DenseNet121, which achieved an accuracy of 90.3% and a specificity of 94.24%. The ensemble of DenseNet121 and MobileNetV2 delivered the best performance, with an

accuracy of 91.33% and a specificity of 96.14%. In contrast, the ViT model underperformed compared to the traditional CNN models, achieving an accuracy of 88.84%. The dataset for the diffuse glioma patch classifier exhibits a significant imbalance, with data from 21 glioma patients and 4 normal patients. This disparity is further amplified by the patch distribution, where glioma patients contribute approximately 8,000 patches and normal patients account for around 29,000 patches. Notably, over 25,000 of these normal patches originate from a single patient, resulting in reduced variability within the normal class. This skewed distribution introduces bias into the dataset, potentially affecting the model's performance and evaluation metrics by over-representing the normal class and limiting feature diversity. Despite these challenges, our proposed ensemble model

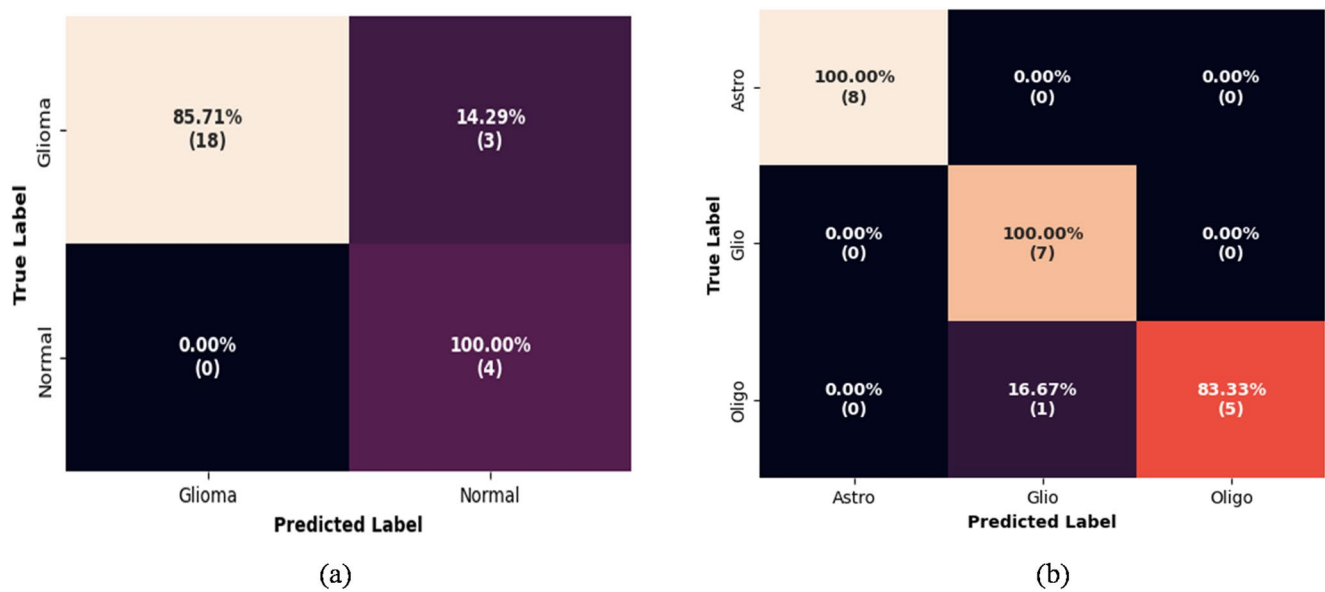


Fig. 10 Confusion Matrix of the Patient-level Classifier **(a)** Diffuse Glioma Classifier and **(b)** Diffuse Glioma Subtype Classifier

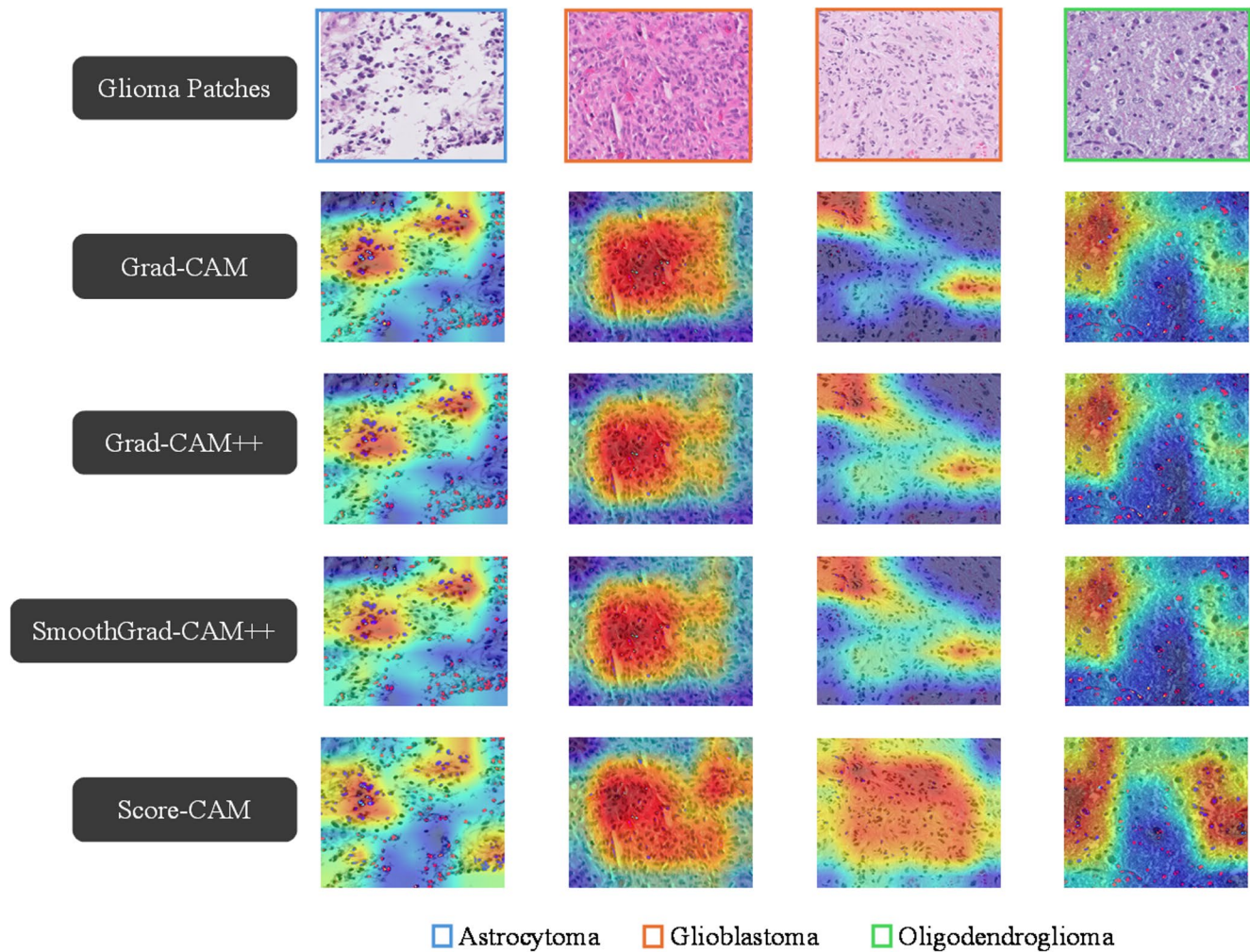


Fig. 11 Visualization of Diffuse Glioma Patches using Explainable AI (XAI) Techniques

demonstrated robust performance, effectively addressing the dataset's limitations.

In our second experiment with the diffuse glioma subtype classifier, we demonstrated the effectiveness of our proposed data preprocessing techniques in addressing the challenges of classifying H&E-stained WSI patches. We evaluated the performance using raw data and the two best-performing normalization techniques—Macenko and Reinhard—alone. Using our proposed ensemble model with raw or unnormalized data, we achieved an accuracy of 58.99%. With data normalized by the Macenko and Reinhard methods, accuracies improved to 66% and 71.47%, respectively. However, these results still did not meet the standards required for medical application. On the other hand, using our novel data preprocessing technique with the same ensemble model, we achieved significant improvement, reaching an accuracy of 89.75% and a specificity of 92.26%. In the preprocessed data, DenseNet121 achieved an accuracy of 90.11%, MobileNetV2 reached 88.02%, and ViT recorded a slightly lower accuracy of 82.77%. In our patient-level classifier, we implemented majority voting based on the results from our best-performing model. For the diffuse glioma classifier at the patient level, we achieved an F1-score of 93.53%. For the diffuse glioma subtype classifier, we recorded a specificity of 97.61% and an F1-score of 95.18%. Despite the challenges posed by dataset variability and staining complexity, the results are robust and reliable, significantly enhancing the clinical-level detection of diffuse glioma.

An important observation from the results of our classifiers is that traditional CNNs and ensemble models outperformed ViT. Conventionally, it is presumed that ViT exhibits superior performance compared to CNNs; however, the efficacy of a model largely depends on the dataset's specific characteristics. Notably, we trained multiple ViT models and have reported only the results from the best-performing model. In our dataset, networks with a comparatively shallower architecture demonstrated enhanced performance, suggesting that the depth of ViT's layers may have been a disadvantage. This finding highlights the importance of adapting the model architecture to fit the unique features of the dataset to optimize performance.

Most of the earlier research in digital pathology for glioma grading has predominantly focused on classifying low-grade gliomas (grades II and III) and high-grade gliomas (grade IV) (Pei et al. 2021; Rathore et al. 2020). However, limited attention has been given to the classification of diffuse gliomas, leaving this area relatively underexplored. Although no prior research has utilized the specific dataset or experimental setup employed in this study, we present a comparison with existing glioma classification studies to offer a comprehensive overview. This comparison underscores the advances introduced by our approach

and its potential to surpass current state-of-the-art methods. Jin et al. (Jin et al. 2021) utilized a dataset of 97,252 histopathological images from 323 glioma patients to classify six categories: oligodendroglioma, anaplastic oligodendroglioma, astrocytoma, anaplastic astrocytoma, glioblastoma, and nontumor images. They proposed SD-Net_WCE, a DenseNet-based model incorporating squeeze-and-excitation blocks and a weighted cross-entropy loss, achieving patch-level accuracy of 86.5% and WSI-level accuracy of 87%. However, their classification framework does not align with the 2021 WHO standards, which emphasize molecular markers for diffuse glioma categorization, potentially limiting its clinical relevance. In contrast, our study adheres to the WHO 2021 standards and demonstrates superior performance in diffuse glioma and subtype classification, as well as patient-wise accuracy. Kurc et al. (Kurc et al. 2020) investigated astrocytoma and oligodendroglioma classification using datasets from the 2018 CPM Challenge, TCGA, and TCIA repositories. Their approach combined radiology-based analysis, which included CNN-based tumor segmentation and PCA for dimensionality reduction, with pathology-based analysis involving ROI detection, patch segmentation, and filtering via isolation forests. A confidence-based voting mechanism integrated the results, achieving an accuracy of 90%. However, the study lacks the explainability, which is essential for interpreting model predictions and aligning them with clinical decision-making. It also does not incorporate patient-wise identification, and the classification performance leaves room for improvement. To address these gaps, our work employs heatmap-based visualization techniques to enhance interpretability and incorporate patient-wise identification, achieving improved classification accuracy. Despotovic et al. (Despotovic et al. 2024) investigated diffuse glioma classification using a dataset of 75 H&E-stained WSIs divided into five classes: astrocytoma, oligodendroglioma, glioblastoma, normal brain tissue, and necrosis. Their out-of-domain pre-trained models, including ResNet50-ViT, achieved a classification accuracy of 94.87%. However, the study lacks the explainability, leaving the model's decision-making process unexamined. Additionally, they did not explore patient-level or WSI-level classification, which is critical for clinical applicability. The use of semi-supervised techniques is also questionable, as these methods are designed for datasets with limited labeled data, making their application to a fully labeled dataset less effective. Table 2 presents a comparison of our work with existing research studies.

While our study addresses key limitations of previous research and makes significant contributions to this domain, it is not without its challenges. The dataset we used lacks variability, particularly in the normal class, which impacted the model's ability to generalize and has led to

Table 2 Comparison of the proposed results with previously published studies

Author	Dataset	Methodology	Result
Jin et al. (Jin et al. 2021)	97,252 histopathological images from 323 glioma patients.	Squeeze-and-Excitation DenseNet (SD-Net_WCE)	Multiclass Classification Accuracy: Patch-level: 86.5%, WSI-level: 87%
Kurc et al. (Kurc et al. 2020)	2018 CPM Challenge (TCGA and TCIA), training set: 32 cases, test set: 20 cases	Radiology-based: Classification using a 3D CNN Pathology-based: Classification using DenseNet Combined prediction: Confidence-based voting method	Combined Radiology and Pathology Image Classification Accuracy: 90% Dropout-Enabled Ensemble Learning Accuracy: 80% Weighted Average-Based Classification Method Accuracy: 75%
Despotovic et al. (Despotovic et al. 2024)	75 H&E-stained WSIs of 29 diffuse glioma cases	Used out-of-domain, in-domain, and semi-supervised transfer learning approaches with ResNet50 + ViT as the model	Out-of-domain Accuracy: 94.87%, in-domain transfer learning Accuracy: 96.39%, and semi-supervised transfer learning Accuracy: 96.91%
Proposed Study	38,544 patches obtained from 25 diffuse glioma patients	Proposed Classification Model: Ensemble of DenseNet121 and MobileNetV2 with majority voting for patient-level classification.	Diffuse Glioma Patch Classifier: Accuracy: 91.33%, Specificity: 96.14% Diffuse Glioma Subtype Classifier: Accuracy: 89.75%, Specificity: 92.26% Patient-Level Classifier (Diffuse Glioma): F1-score: 93.53% Patient-Level Classifier (Diffuse Glioma Subtype): Accuracy: 95.23%, Specificity: 97.61%

misclassifications in this category. Future efforts could benefit from utilizing more diverse datasets with a better representation across all classes. Moreover, incorporating advanced machine learning models and integrating clinical or molecular data could further improve performance and strengthen the model's clinical relevance. One possible approach for integrating molecular data, such as IDH mutation status, is to extract image features and molecular or clinical features separately and then combine them in a late-fusion framework before the final classification layer.

5 Limitation

This study is subject to several limitations and technical challenges that should be carefully considered. The dataset used is notably imbalanced, with disproportionately more patches from normal tissues, especially from a single patient, leading to reduced intra-class variability and potential model bias. This imbalance can affect both the training process and the reliability of performance metrics. Additionally, variations in staining protocols and tissue quality across whole slide images introduce significant challenges for model consistency and generalizability. While our proposed preprocessing technique mitigates some of these issues, it may not fully account for staining artifacts or rare morphological variations. Furthermore, the Vision Transformer (ViT) model underperformed in comparison to CNN-based architectures, indicating that deeper or more complex models may not always be suited for datasets with limited diversity or smaller sample sizes. We also acknowledge that while the visualizations were qualitatively validated by experts, a quantitative agreement measure (e.g., Cohen's kappa) was not conducted. Including such a metric could further support the clinical utility of the explainability techniques and will be considered in future work.

6 Conclusion

This study presents a novel framework for predicting diffuse glioma and its subtypes using H&E-stained whole slide images (WSIs). The methodology is structured into three key stages: a diffuse glioma patch classifier, a diffuse glioma subtype classifier, and a patient-level classifier. A novel data preprocessing technique is introduced to enhance model performance, along with the use of XAI techniques to generate heatmaps that visualize diffuse glioma-affected areas and improve interpretability. This approach addresses key challenges in histopathology image classification, including improving interpretability, managing staining variability, and advancing clinical applications. While the study demonstrates promising results, future efforts will focus on improving generalizability by incorporating more diverse datasets from varied institutions. By leveraging AI-driven methods, the proposed framework enhances diagnostic accuracy, streamlines the diagnostic process, and ensures consistency in clinical assessments, contributing to advancements in precision medicine.

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Authors contribution **Ishrat Jahan:** Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing.

Rafif Mahmood Al Saady: Conceptualization, Methodology, Supervision Validation, Methodology, Software, Writing – review & editing. **Semir Vranic:** Validation, Methodology, Software, Writing – review & editing. **Muhammad E.H. Chowdhury:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing.

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Data availability The processed dataset used in this study can be made available upon a reasonable request to the corresponding author.

Declarations

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Informed consent Statement Not applicable.

Conflict of interest Authors have no conflicts of interest to declare.

Clinical trial number Not applicable.

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